GASTROINTESTINAL DISEASES OF HERPTILES

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UCDAVIS VETERINARY MEDICINE

EXCEPTIONAL CARE, EDUCATION *and* INNOVATION

OUTLINE

Upper GI	Lower GI
• Beak disorders • Stomatitis • Periodontal disease • Infectious disease	• Infectious disease • Foreign body • Constipation • Neoplasia • Cloacal prolapse

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Review your anatomy notes!

ANOREXIA

- One of the most common clinical signs
  - Most often due to poor husbandry
  - Physiologic
- Usually non-specific
- Full husbandry history imperative!
  - Diet
  - Housing – light, humidity, heat support
  - Handling
  - Other pets

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Anorexia is one of the most common presenting complaint for reptiles but is a very non-specific sign that can be related to almost any disease process as well as normal behavioral/physiologic changes. For example, seasonal anorexia is seen in tortoise species that undergo brumation in suitable climates (decreased temperatures and shortened light cycle). Additionally, anorexia or at least hyporexia is very common in species during reproduction. Some species feed less often (refer to nutritional lecture) and the definition of prolonged or sustained anorexia will differ based upon the normal feeding cycle of that animal. For example many snakes feed once a week, so a two week history of anorexia may be less significant than a leopard gecko that had not eaten for that same duration of time.

UPPER GI TRACT

BEAK DISORDERS

- Chelonians- toothless jaw
  - Keratin ridges – rhamphotheca
- Etiology
  - Traumatic
  - Congenital
  - Nutritional deficiencies
    - Ca, improper protein? lack of UVB
  - Infection – bacterial, fungal

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Keratin covers the maxillary and mandibular crests. Beak-like keratin ridges (rhampotheca) along the rostral and lateral aspects of the rhinotheca and gnathotheca are developed. Certain herbivorous species such as the Sulcata have additional keratin ridges present. Historical or acute trauma can cause malocclusion, especially if the underlying bone is affected causing abnormal continued growth. Nutritional deficiencies such as calcium and UVB may effect bone growth. In severe cases of nutritional secondary hyperparathyroidism, bone loss may be seen. Infections can occur primarily but more often secondarily occur due to overgrowth which sets up an ideal environment for bacterial or fungal colonization.

BEAK DISORDERS

- Diagnosis
  - Radiographs – evaluate osteomyelitis
  - Biopsy
  - Culture
- Treatment
  - Debridement
  - Occlusal adjustment
  - +/- topical/systemic therapy

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A diagnostic plan should include at a minimum radiographs to evaluate the severity of the underlying disease. Biopsy can help determine is there is inflammation and microorganisms present. The biopsy may include only keratin or should contain bone if osteomyelitis is suspected. Ideally a culture and biopsy would be performed together as culture alone may sample surface contaminants. Additionally, a culture should be performed on a biopsy sample over a surface sample. Treatment should include removal of infected material. This should be performed under anesthesia or deep sedation, especially if bone involvement is suspected. The beak should be adjusted to try to correct the occlusal plane as much as possible to slow down overgrowth and correct malocclusion. Depending on the underlying disease, topical and or systemic antimicrobials and antifungals may be required.

O'Malley 2008

PERIODONTAL DISEASE

- Inflammation of the soft and hard tissues that support the teeth
- Lizards with acrodont teeth are predisposed
  - Ex: Bearded dragon, Asian water monitors, old world chameleons
  - Teeth are not replaced
  - Thin gingival tissue laterally, attached directly to bone

Acrodont teeth have thin and fragile gingival tissue attaching the tooth directly to the bone of the jaw. The teeth contain dentin ankylosed to the crest of bone of the mandible or maxilla with a central pulp cavity and enamel covering the dentin. When the gingival tissue is disrupted due to trauma, secondary infections can occur.

The gingival margins do not reach the maxillary and mandibular crests, leading to exposure of the bone surface. The gingival margin attaches close to the crest similar to mammals with no periodontal bone exposure. Herps with pleurodont dentition include iguanids, geckos, basilisks and chuckwallas. Thecodont teeth were found in dinosaurs and are not found in extant herptiles.

PERIODONTAL DISEASE

- Causes
  - Age
  - Captive diets – less abrasive
  - Nutritional deficiencies
- Disease
  - Associated with aerobic, anaerobic, Gram's positive, negative infections
    - Severe infections may contain spirochetes, fungi
      - Prognosis guarded with osteomyelitis

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The etiology of periodontal disease is likely multifactorial. In addition to the previously discussed anatomic variants that predispose animals the following are also hypothesized to play a role. With increasing age there is increased tooth wear and increased time for infection and inflammation to occur, especially because these teeth are not replaced. Captive diets may be less abrasive than natural diets. Diets high in soft items such as waxworms may lead to increased plaque build-up and gingivitis. Nutritional deficiencies can also lead to weakening of the epithelial tissues (vitamin A) and bony abnormalities (nutritional secondary hyperparathyroidism). Once gingivitis occurs, there can be loss of periodontal structures and osteomyelitis and/or bone loss. Plaque also sets up an ideal environment for anaerobic infections and spirochetes.

Photo credit: Dr. Paula Rodriguez

PERIODONTAL DISEASE

- Diagnosis
  - Radiographs
  - Anesthetized dental probing
  - Culture
- Treatment
  - Dental cleanings with plaque removal
  - Topical chlorhexidine varnish, doxirobe
  - Antimicrobial therapy
- Prevention
  - Prophylactic treatment every 6-12 months
  - Dietary adjustment – fibrous vegetables, hard-bodied insects

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Diagnosis may include radiographs to determine whether the disease is confined to the teeth and periodontal structures or has progressed to osteomyelitis of the jaw. Depending on the size of the animal this can be done with a standard radiology machine or dental radiographs can be taken for smaller patients. Under anesthesia or sedation, the periodontal pocket can also be probed for any defects similar to what is done in dogs and cats. If significant infection is suspected, a culture should be taken after an initial flushing or cleaning with dilute chlorhexidine to prevent oral contamination.

Treatment should include a method of removal of plaque and tartar if present. This is most effectively done with an anesthetized dental cleaning with either a hand-scaler and or an ultrasonic cleaner. The patient should be intubated for this procedure to prevent the risk of aspiration. This can be supplemented with at home swabbing of the teeth with chlorhexidine and should include systemic antimicrobial therapy, especially if osteomyelitis is present.

Prevention is easier than treatment. You can consider regular dental cleanings for high risk individuals. Some owners can lightly brush their teeth at home using a very small toothbrush or at least CTA's to disrupt plaque development. They should ideally be consuming a diet with abrasive surfaces as well to dislodge and prevent plaque buildup.

STOMATITIS

- Inflammation and typically infection of the tissues of the oral cavity
- Snakes > lizards, chelonians
- Can → rhinitis, sinusitis, osteomyelitis, aspiration, ocular disease, otitis
- Systemic illness
- Signs – petechia, exudate, ptyalism, periodontal disease, ulcerations, difficulty prehending food

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This may be a sign of systemic illness. Progression of this can lead to inflammation and infection of surrounding tissues. There is connection between the lacrimal system and the nasal/respiratory system in most reptiles with the oral cavity. This can also indicate septicemia. Clinical signs can be localized to the oral cavity or extend into other systems. In snakes, monitors and chameleons, tongue sheath abscesses can also occur.

STOMATITIS – ETIOLOGY

- Periodontal disease
- Bacterial – mostly Gram's negative
  - Mycobacterium
  - *Devriesea agamarum* – cheilitis
- Fungal infections
  - *Metarhizium viridae, granulomatis*
  - Opportunistic
- Parasitic
  - Nematodes
  - Spirorchid embolization→ ischemia
  - Flukes

Photo credit: M Roy

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Devriesea sp. is a gram positive rod that most often causes dermatitis. It is a facultatively pathogenic bacteria that can be isolated from healthy bearded dragon oral cavities. However it can cause proliferative hyperkeratotic dermatitis, chelitis and septicemia.

Metarhizium viridae, granulomatis is reported in veiled and carpet chameleons and bearded dragons and can cause glossitis and stomatitis. This fungus is a primary pathogen. Other opportunistic pathogens can also cause infections

This image is of a snake with a mycobacterium lesion invading into the maxilla and surrounding tissues.
Flukes (monogentic and digentic trematode) species parasitize the oral cavity of various reptile species and can cause or exacerbate existing stomatitis, though usually the presence of these parasites is of little clinical concern to the host.

STOMATITIS – ETIOLOGY

- Viral Disease
  - Many cause other organ systems to be affected
  - Herpesvirus – chelonians, lizards; respiratory
  - Paramyxovirus – snakes; respiratory
  - Reptarenavirus – snakes; neurologic
  - Virus X – tortoises; systemic
  - West Nile Virus – crocodiles; neurologic
  - Nidovirus – snakes, lizards; respiratory

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As stomatitis can be an indicator of systemic disease or a diffuse inflammatory condition, many viruses may induce stomatitis as a clinical signs associated with a larger disease process. The following diseases mainly effect other organ systems and will be discussed more in depth in other lectures.

Virus X- Picornaviridae has mostly been reported in spur thighed tortoises.

HERPESVIRUS

- Alphaherpesvirus
- Chelonians
  - Clinical signs – rhinitis, conjunctivitis, stomatitis,
  - Necrotizing glossitis, tracheitis, esophagitis
  - Testudid HV 1-4, Terrapene HV 1
- Lizards
  - Varanid HV 1- Emerald tree monitor proliferative stomatitis
  - Predilection → oral squamous cell carcinoma in green tree monitors
  - Gerrhosaurus herpesvirus 1-2 – stomatitis/chelitis in plated lizards

Mader 2019

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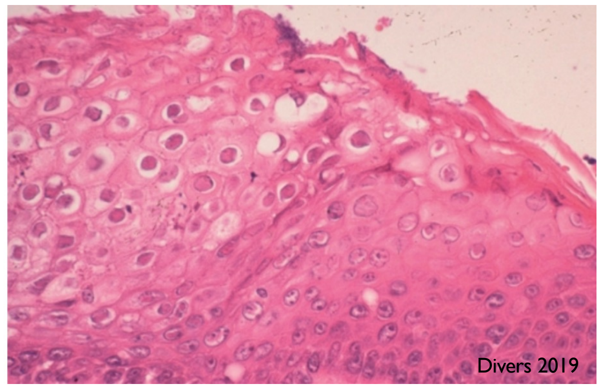
There are many herpesviruses affecting chelonians and squamates. They are significant pathogens of chelonians causing ulcerative signs as well as respiratory illness. Concurrent neurologic, hepatic and splenic disease may be seen based upon the species.
Image: From Mader's Reptile Medicine and Surgery 2019

HERPESVIRUS

- Diagnosis
  - Clinical signs
  - PCR – oral swabs, nasal flushes
  - Serology – latent infections
  - Biopsy with histopathology
    - Eosinophilic intranuclear inclusions
- Treatment
  - Preventative medicine, quarantine
  - Supportive care
  - Antivirals – Acyclovir, lysine?
  - Temperature modulation



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Serology has been performed for tortoises and sea turtles but is not widely available. In sea turtles, serology did not well correlated with disease.

Acyclovir is a nucleoside analogue and the most used antiviral in reptile medicine. It has mainly been used for herpesvirus. It has been shown to inhibit the growth of testudinid herpesvirus 3 in cell culture. Topical and oral therapy has been used. Lysine has been used in tortoises, but efficacy studies are not available and as they stopped using it in cats with herpesvirus, we've gone back to using acyclovir alone for treatment along with supportive care.

Reptiles kept at suboptimal temperatures may be more immunosuppressed than others. Viral replication is reduced at higher temperatures.
Image: From Mader's Reptile Medicine and Surgery 2019. Herpesvirus in a Herman's tortoise- ballooning degeneration of the epithelial cells of the tongue with intranuclear inclusions

RANAVIRUS

- Iridoviridae – Frog virus 3, undescribed
- Chelonians – necrotizing stomatitis, esophagitis, nasal discharge, conjunctivitis, pneumonia, diarrhea, hepatitis, splenitis- DDX herpesvirus, mycoplasma
- Snakes – necrotizing pharyngeal tissue, hepatic necrosis, nasal ulcerations
- Lizards – skin lesions most often
- Diagnosis
  - PCR – oral, cloaca, blood, gastrointestinal tract, liver
  - Serology
  - Intracytoplasmic basophilic inclusions uncommon

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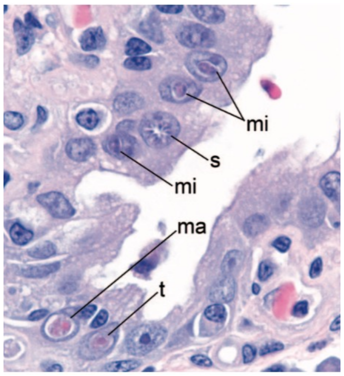
Ranaviruses are from the family Iridioviridae which are DNA viruses. These are a major pathogen in amphibians and have accounted for an enormous number of wild mortality and extinction of several species. However several also affect reptiles, mainly chelonians. Koch’s postulates have been fulfilled for box turtles and red-eared sliders. Most outbreaks have high morbidity and mortality in chelonians.

For PCR it is recommended to combine oral/cloacal swabs and blood for sampling. An ELISA has been developed for Burmese star tortoises, gopher tortoises and eastern box turtles. However antibody levels are typically low. Inclusions are uncommon and challenging to identify

Differentials in most chelonians can be herpesvirus and mycoplasma

INTRANUCLEAR COCCIDIA OF TORTOISES (TINC)



Stilwell JZWM 2017

- **Major emerging pathogen of tortoises**
- Eimeria spp
  - Emerging disease of radiated, leopard tortoises especially sensitive
- Life cycle – unknown
  - Fecal oral proven
  - Oocysts require sporulation (3-4 days)
  - Cloacal shedding in some species
  - Multiple life stages have been found

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TINC is a major pathogen of tortoises. It has only been found in tortoises at this time. Old world tortoises, especially radiated and leopard tortoises seem to be particularly sensitive. Hermann's Russian and African spurred tortoises seem to be more resistant to developing clinical signs but still shed oocysts when experimentally infected. Regardless, all testudinae should be considered to be able to be infected until we have more information.

The life cycle is not completely understood. Recently oocysts have been identified on coproscopy in experimentally infected Leopard tortoises and fecal oral transmission has been proven. The organism may be shed cloacally. This organism has so far only been found in managed tortoises not their wild counterparts.

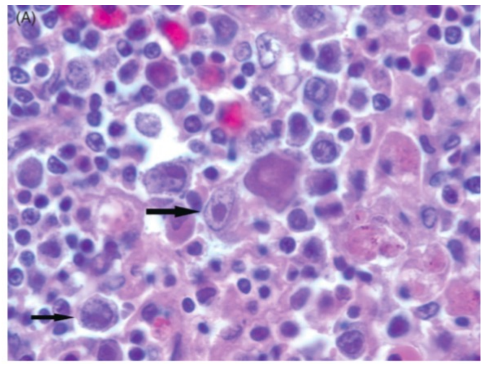
TINC life stages that have been described include trophozoites, meronts, merozoites, macrogametocytes, metocytes and nonsporulated oocysts.

Image: Justin M. Stilwell, Natalie K. Stilwell, Nicole I. Stacy, James F.X. Wellehan Jr., Lisa L. Farina "EXTENSION OF THE KNOWN HOST RANGE OF INTRANUCLEAR COCCIDIOSIS: INFECTION IN THREE CAPTIVE RED-FOOTED TORTOISES (*CHELONOIDIS*

CARBONARIA)," Journal of Zoo and Wildlife Medicine, 48(4), 1165-1171, (1 December 2017). Life stages include microgametocyte (mi), schizont (s), macroga- metocyte (ma) and trophozoite (t).

TINC - DISEASE

- Infects epithelial cells
- Multisystemic disease
- Clinical signs – non-specific
  - Anorexia, lethargy, weight loss
  - Diarrhea
  - Oculonasal discharge
  - GI and oral ulceration/stomatitis
  - Respiratory distress
  - Ascites
- High mortality rates



Terio 2018

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TINC causes a multisystemic disease due to it’s epitheliotropic nature. The clinical signs are therefore nonspecific and may relate to the organ system in which is most affected. Often the time course of disease is rapid with high mortality rates.

Image: Splenic intranuclear coccidia in a star tortoise

TINC

- Diagnosis
  - PCR – nasal, conjunctival, choanal, blood +/- fecal
  - Necropsy and histopathology
- Treatment
  - Ponazuril – improve signs, reduce load
    - Does not eliminate infection-may create carrier state
    - Low oral bioavailability and long half-life in red-footed tortoises (Shemi JZWM 2018)
  - Toltrazuril
  - Prolonged treatment

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Diagnosis is mainly based upon molecular techniques due to the lack of routine shedding of the parasite seen. Different species also shed via different tissues. Nasal swabs, conjunctival swabs, choanal swabs and blood have all been tested for PCR. If testing a species in which the shedding is unknown, multiple tissue samples may be recommended.

Unfortunately at this time there is no effective treatment. Ponazuril improves clinical signs and reduces organism shedding but does not eliminate the infection. It may be more active against actively shedding organisms and not the tissue stage of the infection. A pharmacokinetic study in red-footed tortoises has shown low bioavailability but a long half life. The treatment course should be prolonged (>1 month). Toltrazuril may also reduce clinical signs. As a reminder, ponazuril is the active metabolite of toltrazuril.

There is also the potential for a carrier state where animals clear the infection or test negative and later become positive again.

A PRELIMINARY ANALYSIS OF PROLONGED ABSORPTION RATE OF PONAZURIL IN RED-FOOTED TORTOISES, CHELONOIDIS CARBONARIA Shemi L. Benge, M. Tobias

Heinrichs, Sarah E. Crevasse, Behrang Mahjoub, Charles A. Peloquin, and James F.X. Wellehan Jr. Journal of Zoo and Wildlife Medicine 2018 49:3, 802-805

LOWER GI TRACT

GASTRITIS/ENTERITIS

- Clinical signs - inappetence, diarrhea, constipation, regurgitation/vomiting
- Etiology
  - Infectious – bacterial, viral, parasitic
  - Drug reaction
  - Toxin
  - Foreign body
  - Neoplasia

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Gastritis and enteritis can be caused by a variety of etiologic agents and conditions. Clinical signs associated with them are non-specific and may include anorexia, diarrhea, constipation and in severe cases regurgitation and vomiting. Regurgitation and vomiting are more common in some species such as squamates but is uncommon and more concerning in chelonians. Some regurgitation can be behavioral and may be due to handling for example snakes very soon after eating.

Toxins are overall uncommon but may include toxic plants (common in grazing species or inadvertent feeding of plants by owners) and firefly toxicity (Lucibufagins) in lizards such as the bearded dragon.

BACTERIA

- *Chlamydia pneumoniae* – regurgitation syndrome in emerald tree boas
- Anaerobic infections – *Clostridium*
- *Mycobacterium* enteritis
- *Helicobacter* – enteritis in tortoises
- Treatment
  - Antimicrobial therapy based upon culture
  - Enteral/fluid support

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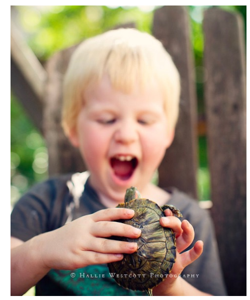
Chlamydia has been associated with a regurgitation syndrome in emerald tree boas. It is a bacterial infection that may cause other organ infections, septicemia and granulomatous disease. This is likely not the sole cause of the syndrome in these snakes but may cause gastric atrophy and fibrosis.

Anaerobic infections may be more common than previously thought. *Clostridium perfringens, C glycolicum, C novyi* have all been reported in reptiles.
Helicobacter has been associated with sepsis secondary to enteritis in several tortoise species

Treatment is based upon the primary etiologic agent. All antimicrobial therapy is ideally based upon culture. However, gastrointestinal culture are challenging in reptiles. Some fecal panels maybe helpful for Clostridial organisms or Chlamydia but contamination and other bacterial overgrowth makes interpretation challenging. Also, the normal bacterial microbiome is not completely understood in reptiles. For example, in the Burmese python the predominant bacteria found during fasting was Bacteroides and the primary bacteria after eating was Lactobacillus. In addition, supportive care including fluid and nutritional support is warranted.

SALMONELLA

- *Salmonella enterica arizonae, S.Typhimurium* are most common
- Consider ALL reptiles carriers!
  - As many as 90% are + on fecal culture
  - Environmental contamination
  - Common in pet reptiles
- Zoonotic
  - Large number of reptile- human cases



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Salmonella is the biggest zoonotic concern with reptiles. Consider all reptiles positive and a potential zoonotic source. It is crucial as veterinarians to include this conversation in the first appointment with every reptile owner to educate them on this risk. There are a large number of reptile to human cases reported. A ban on aquatic turtles <4 inches being sold has led to a decrease in human cases but there is still an ongoing risk. It can also be associated with feeding certain prey items and food can become contaminated as well.

SALMONELLOSIS FROM REPTILES

- Ban on commercial distribution of turtles that are < 4" long in 1975
  - Estimated prevalence of 100,000 cases Salmonella in children per year
  - Illegal distribution continues
  - 10/2013: 473 cases from 43 states, 78 hospitalized
- Pet shops, flea markets, internet, street vendors

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A recent outbreak in 2013 in which 473 people over 43 states were diagnosed with clinical infection, 78 were hospitalized. Source: http://www.cdc.gov/Salmonella/small-turtles-03-12/index.html. Another outbreak in 2015 51 people infected were reported from 16 states, 15 were hospitalized, and no deaths were reported. 50% of ill people were children 5 years of age or younger.

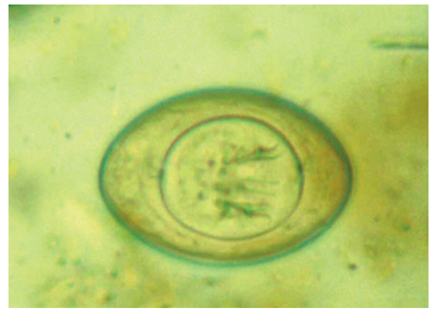
SALMONELLA

- Diagnosis - culture
  - Intermittent shedding
  - What do you do with a positive?
- Treatment
  - Based on culture/sensitivity
  - Avoid fluoroquinolones if possible!!!
- Prevention
  - Hygiene
  - Hand-washing
  - Education

Diagnosis is challenging. This bacteria can be intermittently shed so a single negative culture is not diagnostic. It also becomes challenging as almost all reptiles carry this and should be assumed to be positive. A fecal culture is sensitive or specific for a diagnosis. However an extraintestinal culture such as a blood culture or joint tap is significant and can indicate disease. Treatment should be based upon culture and sensitivity. Because this is a zoonotic disease, antimicrobial resistance is a large concern. Fluoroquinolones are the mainstay to treatment in humans and therefore there use in reptiles should be limited unless absolutely essential to prevent resistant strains from developing. Prevention of contamination of humans is essential. The organizations on the side all have information and handouts on this that can be targeted towards clients. I recommend addressing this during the first visit for any reptile patient and providing your owners with handouts on the risk to them and their families as well as preventative measures that can be taken.

PARASITES

- Not all parasites are pathogenic
  - Some degree is normal
  - Pseduoparasites – prey items
  - Intermediate hosts
  - Commensal/beneficial parasites – herbivores
  - Many flagellates, ciliates - low pathogenicity
- There are many!
  - We will only focus on the most clinically relevant



Wellehan, Divers 2019

UC DAVIS VETERINARY MEDICINE

Most parasite in reptiles are non-pathogenic, especially protozoal parasites which can be a common fecal examination findings. Some degree may even be beneficial or commensal such as in herbivorous tortoises with flagellates. It is also important to differentiate pseudoparasites or parasites form prey items that are just passing through from primary parasites.
Image: *Hymenolepis nana,* pseudoparasite from mouse prey items

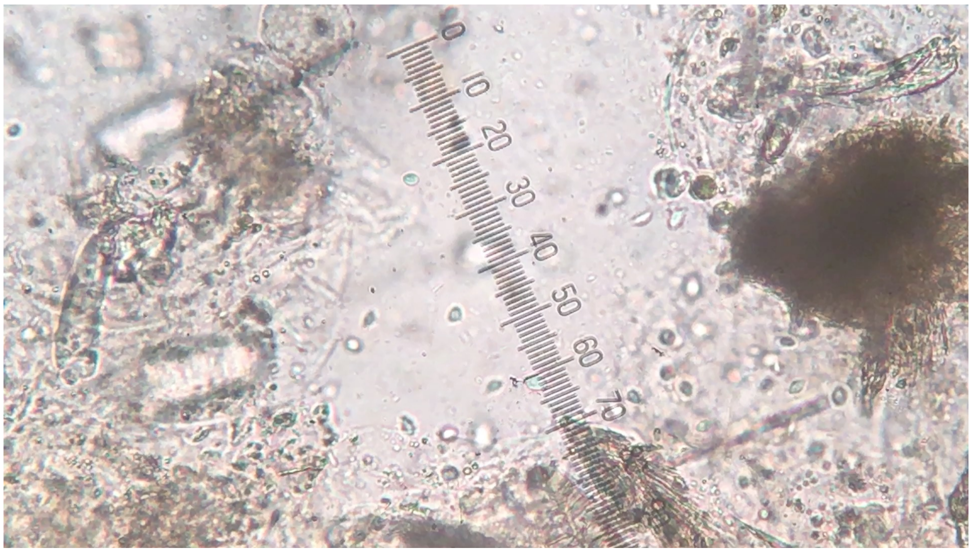
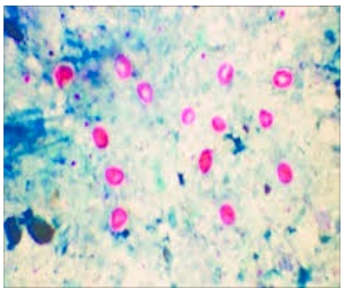


Photo credit: L Camp

This image showed multiple flagellates. These are best visualized on a wet mount within <1 hour of collection while they are still alive. It becomes more challenging to identify these parasites once they are dead and non-motile.

PARASITES – DIAGNOSIS



- Cytologic evaluation
  - Wet mount – motile protozoans
  - Float – coccidia, metazoan eggs
  - Sedimentation – trematodes, acanthocephalons, nematodes
  - Sodium acetate – larval trematodes
  - Modified McMasters – parasite burden
  - Stains – trichrome, Giemsa, acid-fast
- Biopsies
- Immunoassays
- PCR

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Some parasites have high specific gravity and sedimentation may be more sensitive for detection than typical floats. Salt or sugar solution can distort morphologic characteristics. May require feeding in order to obtain a fecal sample or an enema. Alternatively gastric lavages may be submitted based upon the location of interest.

Some parasites, especially those that are closely associated with the luminal surface of gastrointestinal organs are best identified on biopsy and histopathology. In order to speciate organisms, IHC or PCR may be necessary.

OXYURIDS - NEMATODES

- Pinworms
- Direct life cycle
  - Usually host specific
- Common in turtles and lizards
- Rarely pathogenic
  - May have a beneficial effect in some species
- High burden – anorexia, death post-hibernation
- Treatment questionable and prolonged

Photo credit: L Camp

UCDAVIS VETERINARY MEDICINE

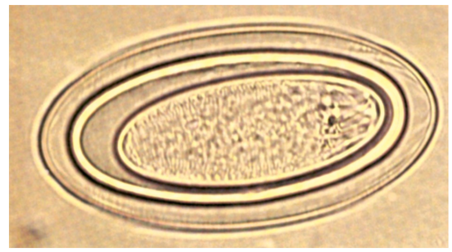
Pinworms are a common fecal examination finding, especially in lizards and turtles. Importantly, they are rarely considered pathogenic. Some have suggested they may be beneficial in some species to help mix fecal material and prevent constipation. However, in high numbers they can lead to anorexia and have been associated with death post-brumation in some tortoises. They have a direct life cycle and therefore environmental cleaning is important in control if this becomes a collection problem. Treatment is challenging and often requires prolonged treatment courses to decrease load. Treatment is not recommended unless loads are very high or it is suspected this is contributing to clinical disease.

Oxyurids are generally thin walled and may have an oval or asymmetrical shape. This is differentiated from ascarid eggs which are usually more round and thick shelled.

Image: Pinworm ova in a bearded dragon

ACANTHOCEPHALANS - NEMATODE

- Spiny headed worms
- Clinical in snakes
- Subclinical in aquatic turtles
- Hematochezia, peritonitis, skin nodules
- Indirect life cycle
  - Invertebrate→ fish/amphibian intermediate host
  - Adults in the small intestines
- Diagnosis – fecal-eggs, multi-enveloped, larva with rostral hooklets
- Treatment - levamisole, loperamide, surgical removal



F Fry, Divers 2019

UCDAVIS VETERINARY MEDICINE

Acanthocephalans are also nematodes. They are typically referred to as spiny or thorny headed worms. They mainly cause disease in snakes and are often subclinical in aquatic turtles. They have an indirect lifecycle that involves an invertebrate and fish or amphibian. The eggs appears multi-enveloped and if you look closely rostral hooklets can be seen within the egg.

Image: Acanthocephalon egg from a monitor lizard

STRONGYLOIDES - NEMATODE

- Significant enteritis in colubrid snakes with *S ophidiae*
  - Diarrhea, weight loss, enteritis
- Life cycle – host and free-living stage
  - Infective stage→ skin → intestines → embeds in mucosa
- Diagnosis – larvated eggs
- Treatment
  - Fenbendazole
  - Ivermectin may cause an inflammatory response?

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Strongyloides are another type of nematode infection. They can cause gastroenteritis. Some species, specifically *Rhabdias sp.* are lungworms that may be found in the feces but primarily cause a respiratory infection. Their life cycle alternates between a host and free-living stage. Larvae or larvated eggs can be seen on fecal examination with a very thin wall.

This infection is mainly seen to cause clinical signs in snakes.

Treatment should be done with caution. Fenbendazole can cause neurological signs in some snakes so dose it low over several days as opposed to high over a shorter period. Snakes that have been treated with ivermectin have died following treatment- suspected to be secondary to a severe inflammatory or anaphylactic reaction from worm death.

NEMATODE TREATMENT

- Ivermectin
  - **DO NOT USE IN CHELONIANS!**
  - Caution in skinks and indigo snakes
  - May be best to avoid altogether!
- Milbemycin oxime
- Fenbendazole – leukopenia can occur
- Levamisole
- Emodepside/praziquantel (Profender®)

Mans 2013

UCDAVIS VETERINARY MEDICINE

The below treatments can also be considered for the treatment of nematodes. However, ivermectin has been associated with deaths in several species. It is absolutely contraindicated in chelonians and can cause respiratory failure and death. Death has also been reported in skinks and indigo snakes with treatment.

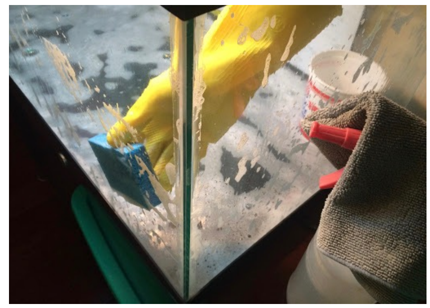
Milbemycin has minimal side effects and has been used effectively in red-eared sliders and box turtles.

Profender is a topical drug that penetrates the skin and is effective against many nematodes. Dosing depends on thickness of the skin and scale type. The most effective location for chelonians is around the axilla or thoracic inlet, snakes near the gular region or cervical spine, lizards in the axilla/neck folds. Avoid letting the reptile soak for 48 hours after treatment. Should be reapplied after 14 days.

Clinical Update on Diagnosis and Management of Disorders of the Digestive System of Reptiles. Christoph Mans. JEPM 2013;22(2):141-162

PARASITE PREVENTION

- Quarantine
  - Testing
- Removal of feces from the cage
- Daily cleaning
- Use easily cleanable cage furniture
- Easily cleanable substrate
- Remember – direct life cycle = reinfection



UCDAVIS VETERINARY MEDICINE

Reptiles entering a collection should always be quarantine for a minimum of 30 days with some institutions preferring 60 days in total. Quarantine means a separate room, ventilation system and no direct handing from collection animals to quarantine animals. Many owners do not practice this or have breaches in their quarantine system that can still lead to contamination between animals. For species with a direct life cycle, environmental cleaning is essential as auto-infection is very common. In general, care must be taken with cleaning to ensure that toxic materials do not come in contact with the reptile or with their respiratory system. For a majority of parasites, cleaning with soap and water and rinsing will remove them unless there are porous structures within the enclosure. UV light and heat can also be utilized following this treatment. Other cleaning solutions such as peroxide or bleach based (do not mix the two!) can be used but is recommended to fully rinse out the enclosure after cleaning and allow at least several hours before the animal is placed back in the enclosure to avoid aerosol irritation.

ISOSPORA AMPHIBOLURI - COCCIDIA

Photo credit: L Camp

- Common in young bearded dragons
- **<u>Usually asymptomatic</u>**!
- Enteritis with very high burdens seldom reported
- Signs – seldom but anorexia, lethargy, weight loss, diarrhea, prolapse, death has been reported
- Small and large intestine
- Treatment – sulfadimethoxine, ponazuril, if you treat at all!
  - Microbial stewardship here

UCDAVIS VETERINARY MEDICINE

Coccidia are very common in bearded dragons. Most often they are asymptomatic however enteritis has been seen with high burdens as well as with adenovirus infections. The infection appears to start at the duodenum and progress to the colon. Treatment with sulfa based drugs for a month has reduced shedding but is is usually not fully eliminated.

Image: Singleton B et al. Evaluating Quikon® Med as a Coccidiocide for Inland Bearded Dragons (Pogona vitticeps). Journal of Exotic Pet Medicine, Vol 15, No 4 (October), 2006: pp 269-273

CRYPTOSPORIDIUM - COCCIDIA

Photo credit: K Berg

- Devastating disease
- Worldwide distribution
- Chronic wasting syndrome and maldigestion
- NOT zoonotic
  - r/o pass through parasites from prey items

UCDAVIS VETERINARY MEDICINE

Cryptosporidium is a major parasitic pathogen of reptiles and can be a devastating disease to a collection. There are a variety of species that infect reptiles can the presentation varies between species. However in general, this causes a chronic wasting syndrome secondary to maldigestion of food. It is important to note that the *Cryptosporidium sp.* that infect reptiles are not zoonotic. They can infect other reptiles but not humans. It is also important to differentiate parasite such as *Cryptosporidium muris* from mouse prey in snakes.
Subclass coccidia, family Cryptosporidiidae

CRYPTOSPORIDIUM

Gastric

- *Cryptosporidium serpentis*
- *Cryptosporidium testudines*

Intestinal

- *Cryptosporidium varanii*
- *Cryptosporidium ducismarci*

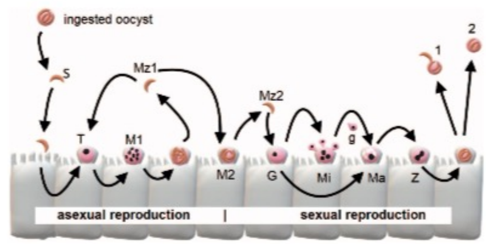
Classically *C serpentis* infects snakes, *C varanii* lizards, and *C testuines and C ducismarca* chelonians. There is some cross over for infectivity between reptile species. For example, *C serpentis* can infect other reptiles. *C ducismarci* has been reported in a marginated tortoise, Russian tortoise, pancake tortoise, ball python and veiled chameleon.

C testudines has been reported in a Russian tortoise.

In general the important part is what portion of the gastrointestinal tract they effect.

CRYPTOSPORIDIUM

- Life cycle – direct
  - Inhabit an extracellular vacuole in the microvillus
  - Infective stage = sporulated oocyst passed in feces
  - Sexual and asexual reproduction
  - Environmentally resistant
- Transmission – fecal-oral



Bogan JHMS 2019

UCDAVIS
VETERINARY MEDICINE

As with many coccidia, this is a direct life-cycle. The complete life-cycle is not known but is likely similar to the mammalian types. The parasite inhabits a vacuole in the microvillous region of the gastrointestinal tract. It is located in a more superficial location rather than deeply invasive. The infective stage is a sporulated oocyst that passes in the feces and is immediately infective. Transmission is fecal-oral. It can persist in the environment for months. Extra-intestinal infections have been reported in the green iguana and it is a cause of aural/pharyngeal polyps.

James E. Bogan Jr (*2019*) Gastric Cryptosporidiosis in Snakes, a Review. Journal of Herpetological Medicine and Surgery: December 2019, Vol. 29, No. 3-4, pp. 71-86.

CRYPTOSPORIDIUM – CLINICAL DISEASE

- Subclinical carriers most common in snakes
- Weight loss, chronic wasting
- Undigested food in feces
- Gastric
  - Snakes- hyperplasia, midbody swelling
    - Regurgitation
  - Lizards -severe enteritis
    - Diarrhea, hematochezia

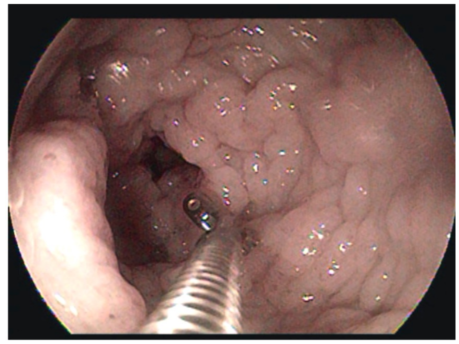
Photo credit: D Mader

UCDAVIS VETERINARY MEDICINE

Clinical signs will vary depending on the species of parasite and the species of reptile infected. Again remembering that some tend to have an intestinal predilection and others gastric. Subclinical carriers are most common in snakes. Can pass oocysts intermittently without clinical disease. In both forms weight loss is a common feature. Once disease becomes severe it is often fatal.

CRYPTOSPORIDIUM – DIAGNOSIS

- Acid fast or IFA serology
  - Sample- gastric wash, regurgitant
  - Increased sensitivity- 3 days after feeding
  - Fecal or cloacal swab from geckos
- Contrast GI study – narrowing gastric lumen
- **Gold standard – gastric biopsy (snake)**
- PCR test – ID species



Bogan JHMS 2019

UC DAVIS VETERINARY MEDICINE

Acid fast and IFA serology will only tell you there is a cryptosporidian species there. It will not differentiate one from a prey item nor which reptilian one is present

Gastric washes are the preferred sample for gastrotrophic species and feces for intestinal species. Testing 3 days after feeding or gavage feeding of snakes has shown to increase the sensitivity or changes of detection.

For snakes, the gold standard is still histopathology by gastric biopsy as this has been found to have higher sensitivity compared to IFA or PCR. PCR is needed to speciate. Antibodies develop at least in snakes following infection, but take at least 6 weeks to develop. There is no longer a commercially available antibody ELISA.

CRYPTOSPORIDIUM - TREATMENT

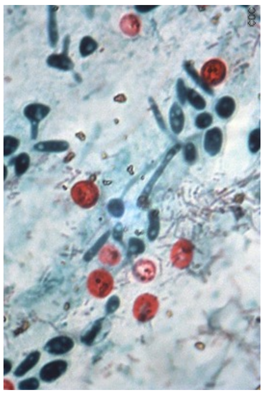


Photo credit: J Paul-Murphy

- Challenging, not self-limiting
- Management
  - Testing, quarantine
  - Consider culling
  - Disinfectant: 6-10% hydrogen peroxide or ammonium hydroxide
- Paromomycin – most promising for 40 years
  - Aminoglycoside– not systemically absorbed
- TMS, nitazoxanide – may reduce signs, does not eliminate
- Hyperimmune bovine colostrum

UCDAVIS VETERINARY MEDICINE

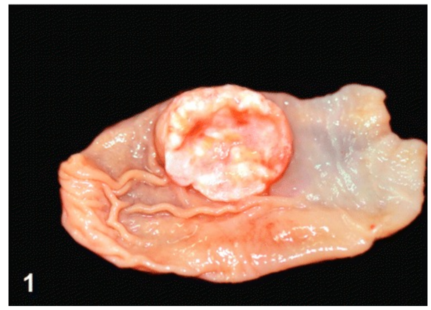
Treatment is often unrewarding and no effective protocols have been established. While this is typically self-limiting in mammals, this is NOT the case for reptiles. Often times this is a management issue as it is readily transmitted to other in the collection and the best efforts may be at eliminating this from a collection. This is done with appropriate quarantine, hygiene and testing. Some collections may consider culling positive animals due to challenges with treatment. Supportive care is an important component to therapy with aggressive enteral nutritional support.
This parasite is resistant to many types of disinfectants. Hydrogen peroxide may be one of the more effective ones and requires 13 min contact time with 6% for reduce infectivity and excellent decontamination with 10% for 2 hours.

The most promising drug is paromomycin. This is an aminoglycoside that is not systemically absorbed from the GI but treats infections within the GI tract. It has been shown to decrease shedding at high dosages in bearded dragon. It may be price limiting for some patients. Side effects such as nephrotoxicity can occur if gastrointestinal ulceration is present allowing for systemic absorption.

Hyperimmune bovine colostrum cleared infection in 50% of snakes. Other studies have shown it reduces shedding in snakes and leopard geckos. Getting this product though is not available commercially but can be available in some zoo settings.

GASTRIC NEUROENDOCRINE TUMORS OF BDS

- Primarily bearded dragons, of all ages
  - Highly malignant - metastasis → liver
  - Mostly somatostatinomas
  - Diagnosis
    - Weight loss, vomiting, melena
    - Bloodwork – **hyperglycemia**, anemia
    - FNA via US
  - Treatment
    - Surgical treatment not yet established for this gastric tumor
  - Prognosis
    - Guarded to grave



Ritter J 2009

UCDAVIS VETERINARY MEDICINE

Gastrointestinal tumors are overall uncommon in reptiles but should be suspected with non-healing ulcers/erosions, recurrent abscessation, or nonspecific inflammatory conditions.

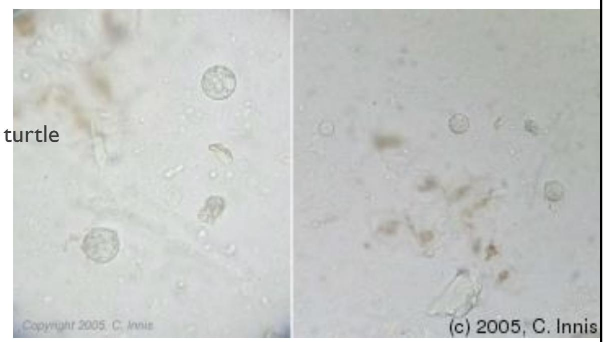
The more common neoplasia reported is most often seen in Bearded dragons. It is a highly malignant gastric neuroendocrine tumor. These tumors arise from the neuroendocrine cells of the mucosa and submucosa of the stomach. The most common site of metastasis is the liver but other commonly reported organs are the kidney, pancreas, intestines and heart. The most often hormone secreted is somatostatin. Classic signs include hyperglycemia, vomiting and melena causing anemia. There is report of diagnosis of this tumor on FNA of the metastatic masses.

Ritter, J. M., Garner, M. M., Chilton, J. A., Jacobson, E. R., & Kiupel, M. (2009). Gastric Neuroendocrine Carcinomas in Bearded Dragons (Pogona vitticeps). *Veterinary Pathology*, *46*(6), 1109–1116.

ENTAMOEBA INVADENS - AMOEBA

- Gastrointestinal parasite of captive snakes
- Lizards – herbivores may be less susceptible
- Chelonians – usually asymptomatic/reservoir
  - Epizootics: red-footed tortoise, leopard , flat-shelled turtle sea turtles
- Mixed exhibits most common for outbreaks

- Clinical Disease
  - Lethargy, depression, regurgitation hematochezia, cloacal prolapse, death
  - Tissue invasion – liver, kidney
  - Primarily enteritis, ulceration of the GI tract

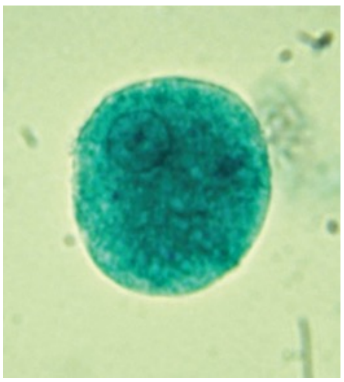


UCDAVIS
VETERINARY MEDICINE

Entamoeba invadens is an important disease in reptiles. This may have been seen more often historically than it is currently but an uptick has occurred in the 2020s.

It is often asymptomatic in chelonians, though disease has been reported in some species. The classic example of an infection is a mixed exhibit with these chelonians and other animals. Snakes may develop more significant and severe disease. This is a primarily gastrointestinal parasite but tissue invasion of the liver and kidney can occur as the disease progresses.

ENTAMOEBA INVADENS - AMOEBA


Divers 2019

- Life cycle
  - Binary fission
  - Ingestion of cyst→ motile trophozoite in large intestine
  - Can spread hematogenously
- Diagnosis
  - Wet mount with trichome staining
  - PCR testing
  - Histopathology

UC DAVIS VETERINARY MEDICINE

This parasite has a direct life cycle and reproduces by binary fission. It has a fecal-oral transmission route. The cysts are ingested and become feeding, motile trophozoites in the large intestine where they begin to reproduce. They may burrow into the mucosa and cause ulcerations in immunosuppressed animals. The infection can spread hematogenously or through the portal vein into the liver and can also cause renal and pulmonary infections. Diagnosis is typically achieved in a wet-mount. It is often challenging to see without additional stains. Carrier animals may be difficult to detect due to low numbers of parasite that are shed. There is a PCR available but it has a low sensitivity and may require multiple samples. On histopathology you may appreciate thickening of the intestines, erosions, ulcerations and abscesses in multiple tissues, namely the liver.

ENTAMOEBA INVADENS - AMOEBA

- Treatment
  - Must target trophozoite and cyst
  - Metronidazole
    - Death in some snakes
  - Paromomycin
  - Temperature manipulation
- Management
  - Avoid mixed collections!!!
  - Environmental disinfection, quarantine

UC DAVIS VETERINARY MEDICINE

Treatment can be challenging as effective treatment but target the cyst as well as the trophozoite form.

Metronidazole at high doses has caused death and neurologic signs in indigo snakes, rattlesnakes and king snakes.

Potentially temperature-sensitive pathologic action (most severe in snakes at 25°C). Snakes at 13, 33 and 35C did not develop disease. May consider increasing temperatures for treatment

FOREIGN BODY

Photo credit: K Berg

- Location – oral cavity, esophagus, stomach, intestines, cloacal
- Causes
  - Substrate
  - Pica
  - Fishing hooks
  - Foreign material
  - Urinary calculi

UCDAVIS VETERINARY MEDICINE

Foreign bodies may be found in any portion of the gastrointestinal tract and they may or may not cause an obstruction based upon the location, type of material and size. The most common foreign body in most of our pet animals is accidental substrate ingestion. For example, consumption of gravel or sand when animals are fed on this material can lead to impaction and secondary obstructions of the GI tract. This is most common in insectivorous lizards and chelonians. It may also occur when snakes are fed on fibrous substrate. Towel or fabric foreign bodies are common in snakes who ingest them while trying to eat a rodent prey in their enclosure. Fishing hooks in the upper gastrointestinal tract occur in aquatic turtles and wildlife. Urinary calculi that lodge within the cloaca can cause a gastrointestinal obstruction.

Smokey Jo 949229

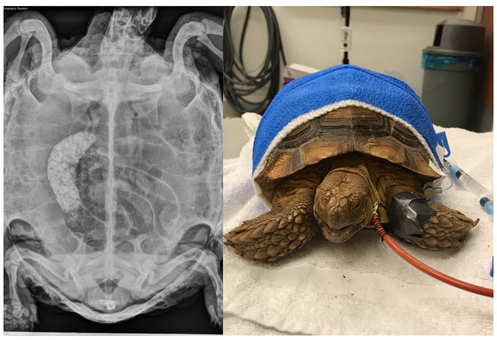
FOREIGN BODY

- Clinical signs- ptyalism, clawing at mouth, inability to swallow, vomiting, regurgitation, anorexia, weight loss, coelomic distension
- Diagnosis
  - Radiographs
  - Contrast radiographs
  - Ultrasound
  - CT scan
  - Endoscopy

UCDAVIS VETERINARY MEDICINE

Plain radiographs may be helpful for radiodense material, however other material such as plastic and wood may be harder to detect unless it is causing an obvious obstructive pattern. The GIT can be visualized well with ultrasound and CT scan if radiographs alone are not diagnostic. Ultrasound and or CT scan be helpful to determine whether perforation or peritonitis is present. Endoscopy can be diagnostic and therapeutic but is only feasible for the upper gastrointestinal tract (oral cavity, esophagus, stomach, potentially proximal duodenum) or cloaca/distal colon.

TREATMENT – MEDICAL



- Non-obstructive
- Supportive feeding
  - Critical care diet
  - Fiber for herbivorous species
  - Esophagostomy tubes – chelonians
- Hydration
  - Subcutaneous
  - Oral

UCDAVIS
VETERINARY MEDICINE

Medical therapy maybe indicated for cases that are not obstructive. This is most common with substrate ingestion or pica. Often times this material if it is going to pass requires fluid therapy as well as oral feeding. This is a common condition in tortoises, especially those that are housed outdoors. Ideal enteric nutritional support should be readily digestible and have a reasonable fiber content, especially for herbivorous species to encourage gastrointestinal motility. Oral feeding of chelonians is aided by a feeding tube which can be used to give mediations, fluids and food. Prior to feeding an animal, it is crucial to ensure they are appropriately hydrated which can be done with oral and percutaneous fluid support.

CONSTIPATION

- Etiology
  - Chronic dehydration
  - Inappropriate temperature
  - Lack of fiber
  - Lack of exercise
  - Hard to pass food items
  - Metabolic
  - Extraluminal compression

Constipation is a common problem in patients that are dehydrated or anorexic. Chronic dehydration may be due to lack of oral intake or environmental humidity. This can also be due to an inappropriate died with increased chitinous material or inappropriate fiber depending on the dietary style of the animal. Metabolic causes such as hypocalcemia and extraluminal compressions of the colon from masses are additional etiologies.

CONSTIPATION

- Diagnosis - radiographs, contrast radiographs
- Treatment
  - Enteral therapy
  - Enemas or cloacal washes
  - Soaking
  - Medical therapy – lactulose, psyllium
  - Vibration
  - Acupuncture

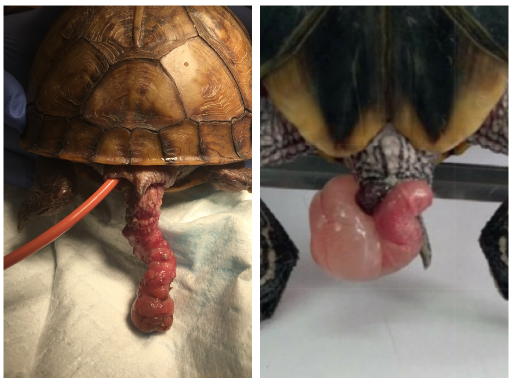
UCDAVIS VETERINARY MEDICINE

Diagnosis of constipation is straightforward and is aided by diagnostic imaging. Keep in mind when performing a contrast study that the transit time will being more prolonged than usual if this is present.

Enteral therapy including both oral fluids and slow re-introduction to feeding is one of the most important components to treatment. The volume and rate of oral feeding will depend on the chronicity of the condition. Additional ways to offer fluid support include soaking and enemas can be performed to help loosen fecal material. The recommended material for enemas is warm water and sterile lubricant that is most easily administered in an awake or sedated patient with a red rubber catheter. This is a blind technique and there is a risk in species with a bladder of entering the wrong location.

There is evidence that vibration therapy may help, especially in chelonians. Acupuncture is also an adjunctive treatment that can be used, however no prospective studies have been performed on it’s efficacy for this condition but we have used it successfully here at UCDavis.

CLOACAL PROLAPSE



Left: Gleeson, Right: Vinkerkar, Divers 2019

- May be related to disease of the gastrointestinal tract, reproductive tract, urinary system or disease of the cloaca
- Cloaca – coprodeum, urodeum, proctodeum
- True emergency!
  - Devitalization can occur
  - Obstructions
- Try to identify the tissue that prolapsed
  - Phallus has best prognosis

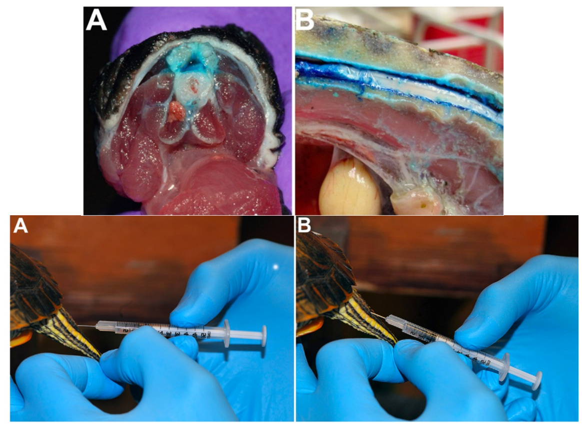
UC DAVIS VETERINARY MEDICINE

This is one of the few true reptile emergencies. With the exception of the male copulatory organs, other prolapses may be fatal as they will become rapidly devitalized and infected. It is important to identify the tissue of origin for these prolapses.

Image: Bladder prolapse, red eared slider
Oviduct will have longitudinal striations present. Endoscopy may be required to help identify the tissue.

CLOACAL PROLAPSE - TREATMENT

- Analgesia!
  - Systemic analgesics and anti-inflammatories
  - Local – topical or intrathecal block
  - Sedation/anesthesia required for reduction
- Stabilization
  - Fluid support
  - Nutritional support
  - Thermal support
  - Treat infection/sepsis



Mans 2014

UCDAVIS VETERINARY MEDICINE

Analgesia is crucial for the treatment of prolapses. This is a painful condition. Additionally, you need to remove or decreased the cloacal tone in order to replace the tissues. Most often full anesthesia is required. Local therapy in the form of an intrathecal block can be performed with similar effects to an epidural in mammals. This is well studied in chelonians. They lack a well developed epidural space but have a well developed intrathecal or subdural space which surrounds the spinal cord. In chelonians you can use this to block coccygeal structures.

The patient must be stabilized prior to anesthesia or surgery if required. Additionally, secondary infections are common with this condition due to possible devitalization of tissue and translocation.

Christoph Mans, CLINICAL TECHNIQUE: Intrathecal Drug Administration in Turtles and Tortoises, Journal of Exotic Pet Medicine, Volume 23, Issue 1,2014, Pages 67-70,
(A) Cross-section of the proximal tail base. The close proximity of the skin overlying the neural arch dorsally can be noted. (B) Sagittal view of the spinal canal at the level of the last dorsal and sacral vertebrae; (A) Penetration of the skin and neural arch with the 28-gauge half-inch needle at approximately 451. (B) Advancement of the needle into the spinal canal at approximately 45-90 degree angle

CLOACAL PROLAPSE – TEMPORARY REDUCTION

- Clean/debride
  - Assess viability; resect any necrotic material
- Osmotic agents- cold, hypertonic saline, dextrose, sugar may be used
- Sterile lubricant +/- analgesic
- Reduce using CTA, swab, fingers
- Once reduced, consider cloacal flushing with warm fluids
- Reduction of vital tissues may require a celiotomy after
- When reduction is not possible→ phallus amputation, celiotomy
- Sutures, staples can support the vent and prevent future prolapse

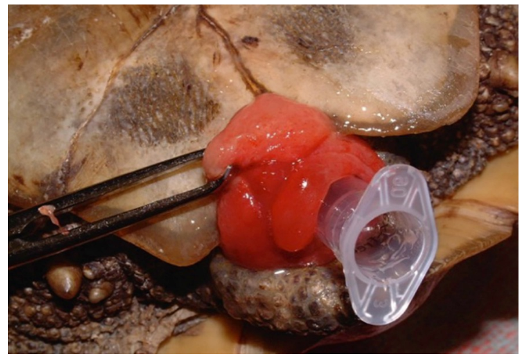
Photo credit: M Gleeson

UCDAVIS VETERINARY MEDICINE

Reduction is only recommend if the tissue appears viable.

CLOACAL PROLAPSE - TREATMENT

- Celiotomy often required
  - Resection, including bladder resection
  - Cloacopexy
- Removal of tissue
  - Resection and anastomosis
  - Reproductive- phallectomy
- Euthanasia
- Prognosis
  - Good for acute phallus prolapse
  - Poor for more chronic prolapse of bladder, GI tract



S McArthur, Divers 2019

UCDAVIS VETERINARY MEDICINE

For many tissues a celiotomy is required either following temporary reduction or alone. The affected tissue may be resected if non-viable. You can resect the full bladder of a reptile if needed but they will be prone to dehydration in the future. Certain tissues such as the colon or cloaca may be pexied to prevent recurrence. If the tissue is non-viable removal is recommended.

Reproductive tissue is non-essential and can be removed. The male phallus is not required for urination and can be removed without a celiotomy. For severe cases or those with gastrointestinal necrosis euthanasia may be considered.

QUESTIONS?